Siegel Zeroes of Dedekind Zeta Functions

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Introduction

These are notes I've written to facilitate my own understanding of Siegel Zeros of Dedekind zeta functions. These are based on lectures given by Dr. Jeffrey Hoffstein for the Spring 2019 Siegel Zero seminar at Brown and notes on analytic number theory he was kind enough to share. All errors are almost certainly mine.

1 The Γ-Function

Definition. *We say that a complex valued function f is* ***complex analytic*** *(or just analytic) over a domain D if for every $x \in D$, there is a neighborhood N of x such that f is differentiable over N.*

Definition (Courtesy of Churchhill & Ruel's text)**.** *Letting $D_1, D_2 \subset \mathbb{C}^n$ with $D_1 \cap D_2 \neq \emptyset$ and f_1 analytic over D_1, then if there exists a function f_2 analytic over D_2 which the additional property*

$$\forall z \in D_1 \cap D_2 : \quad f_1(z) = f_2(z),$$

then f_2 is called the ***analytic continuation*** *of f_1 into the domain D_2.*

Theorem 1. *Analytic continuations are unique.*

Proof. Suppose that f is analytic over D_1 and that f_2 is an analytic continuation of f_1 into D_2 with $D_1 \cap D_2 \neq \emptyset$. Now, suppose there exists another analytic continuation of f_1 into D_2, which we'll call g.

If $g \neq f_2$, it follows that there is some $z \in D_2$ such that $g(z) \neq f_2(z)$. Since g i s analytic, it follows that every sequence $S = (g(z_1), g(z_2), \dots)$

Since g is an analytic continuation of f_1 into D_2, it follows that for every $z \in D_1 \cap D_2$, we have $f_1(z) = g(z)$. ■

Definition. *The Γ-Function is defined as follows for $z \in \mathbb{C}$ with $Re(z) > 0$:*

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$$

So why might we care about this odd looking thing? Basically, it is the analytic continuation of the factorial function, a function that behaves exactly like the factorial function does on positive integers, but is also defined on the entire "right half" of the complex plane.

Proposition 1. *For any $z \in \mathbb{C}$ where $Re(z) > 1$ we have*

$$\Gamma(z+1) = z\Gamma(z)$$

Proof. We recall that

$$\Gamma(z+1) = \int_0^\infty x^z e^{-x} dx$$

integrate by parts with $u = x^z$, $du = zx^{z-1}$, $dv = e^{-x}$, and $v = -e^{-x}$, which gives us

$$\Gamma(z+1) = \int_0^\infty x^z e^{-x} dx = \Big[-x^z e^{-x}\Big]_0^\infty + \int_0^\infty zx^{z-1}e^{-x} dx = 0 + z\int_0^\infty x^{z-1}e^{-x} = z\Gamma(z)$$

■

From the above proposition and a simple inductive argument, it is easy to demonstrate that given any $n \in \mathbb{N}$, we have

$$\Gamma(n+1) = n!$$

Introduction to the Riemann Zeta Function

Definition. *Letting $Re(z) > 1$, the **Riemann-zeta function** is defined to be the Dirichlet series*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^z}.$$

We see that $\zeta(s)$ converges on the right half of the plane past $Re(z) > 1$. We can, however, extend ζ to most of the complex plane via analytic continuation. though the primary one of interest to us is derived from the **functional equation**

$$f(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)f(1-s) \tag{2}$$

Where $\zeta(s)$ is the unique solution f. There is another characterization of the functional equation that will be of more use to us, called the xi-function:

$$\xi(s) = s(s-1)\tfrac{1}{2}\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \tag{3}$$

yielding with a bit of algebra another characterization of the Riemann-zeta function:

$$\zeta(s) = \frac{\xi(s)\pi^{s/2}}{\Gamma(\frac{1}{2}s+1)(s-1)}$$

The reason we are particularly enamoured of this version of the functional equation is that it satisfies a symmetry property

$$\xi(s) = \xi(1-s)$$

Values of ζ(s) on and near the critical strip

Introducing some new notation, the "less-than-less-than" symbol, whose meaning is defined as follows:

$$f(z) \ll g(z) := \exists c : \ \exists z_0 : \ \forall |z| > |z_0| : \ |f(z)| < c|g(z)|$$

By the Phragmén-Lindelöf and Convexity Principles, we have that

$$\zeta(1 + \epsilon + it) \ll_\epsilon 1 \tag{4}$$

where $\ll_\epsilon$ means that the required constant c alluded to in the definition of $\ll$ depends on ϵ. This gives a bound on ζ to the right of the critical strip. It is reasonable to ask if there is a useful bound on ζ on the other side of the critical strip. The answer is yes and the proof involves some clever manipulation of the functional equation.

The symmetry property of ξ along with the characterization of ξ given by Equation 3 tells us that

$$\xi(s) = s(1-s)\pi^{-\frac{s}{2}}\Gamma(\tfrac{s}{2})\zeta(s) = s(1-s)\pi^{-\frac{1-s}{2}}\Gamma(\tfrac{1-s}{2})\zeta(1-s) \tag{5}$$

By dividing everything by $s(1-s)\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$ we derive

$$\zeta(s) = \pi^{s-\frac{1}{2}}\zeta(1-s)\frac{\Gamma(\frac{1-s}{2})}{\Gamma(\frac{s}{2})} \tag{6}$$

Setting $s = -\epsilon + it$, we get

$$\zeta(-\epsilon + it) = \pi^{-\epsilon+it+\frac{1}{2}}\zeta(1+\epsilon-it)\frac{\Gamma(\frac{1+\epsilon-it}{2})}{\Gamma(\frac{-\epsilon+it}{2})} \tag{7}$$

Since $|\zeta(s)| = |\zeta(\overline{s})|$, we know that $\zeta(1+\epsilon-it) = \zeta(1+\epsilon+it)$. Furthermore $|\pi^{-\epsilon+it+\frac{1}{2}}| = |\pi^{-\epsilon+\frac{1}{2}}|$ is just some constant depending on ϵ. It follows then from Equation 4 that

$$\zeta(-\epsilon + it) \ll_\epsilon \frac{\Gamma(\frac{1+\epsilon-it}{2})}{\Gamma(\frac{-\epsilon+it}{2})} \tag{8}$$

There is a useful classical bound that we will exploit for our purposes:

Proposition 2 (Stirling's Approximation formula)**.**

$$\Gamma(s) = \sqrt{\tfrac{2\pi}{s}}(\tfrac{s}{e})^s(1 + O(\tfrac{1}{s}))$$

where $O(\frac{1}{s})$ is a function such that there is a real constant K such that $|O(\frac{1}{s})| \leqslant K|\frac{1}{s}|$. More information can be found in De Bruijn's Asymptotic Methods in Analysis.

Proof. Here's a proof ■

Setting $s = \sigma + it$, we fix σ and let $t \to \infty$. We acquire a bound on the modulus:

$$|\Gamma(s)| = \left|\left(\frac{s}{e}\right)^s \sqrt{\tfrac{2\pi}{s}}\left(1+O(\tfrac{1}{s})\right)\right| \ll \left|\frac{\sqrt{2\pi}s^{s-\frac{1}{2}}}{e^s}\right| \cdot \left(1+|\tfrac{1}{s}|\right) = \left|\frac{\sqrt{2\pi}e^{(\sigma log(s)+itlog(s))}s^{-\frac{1}{2}}}{e^s}\right|\left(1+|\tfrac{1}{s}|\right)$$

Using the exponential form $s = re^{i\theta}$, we compute $log(s) = log(r) + i\theta$ and observe that as $|t| \to \infty$, $\theta \to \pm\frac{\pi}{2}$. After noting $r = |r| = \frac{|t|}{|sin(\theta)|} \to |t|$. Making the relevant substitutions and simplifying yields

$$\sqrt{2\pi}e^{\sigma log|t| - sign(t)\frac{t\pi}{2}}\frac{|t|^{-\frac{1}{2}}}{e^\sigma} = \frac{\sqrt{2\pi}}{e^\sigma}|t|^{\sigma-\frac{1}{2}}e^{-\frac{\pi}{2}|t|}$$

Giving us a modified version of Stirling's formula:

$$\boxed{|\Gamma(\sigma + it)| \ll |t|^{\sigma-\frac{1}{2}}e^{-\frac{\pi}{2}|t|}} \tag{9}$$

With this in hand, we return to our work from Equation 8 and compute

$$\frac{\Gamma(\frac{1+\epsilon-it}{2})}{\Gamma(\frac{-\epsilon+it}{2})} \ll_\epsilon \frac{|\frac{t}{2}|^{\frac{1+\epsilon}{2}-\frac{1}{2}}}{|\frac{t}{2}|^{\frac{-\epsilon}{2}-\frac{1}{2}}}e^{-\frac{\pi}{2}(\frac{|t|}{2}-\frac{|t|}{2})} = \left|\frac{t}{2}\right|^{\frac{1}{2}+\epsilon} \ll_\epsilon |t|^{\frac{1}{2}+\epsilon}$$

giving us a bound on values of $\zeta(s)$ "immediately to the left" of the critical strip:

$$\zeta(-\epsilon + it) \ll_\epsilon |t|^{\frac{1}{2}+\epsilon} \tag{10}$$

Between Equation 10 and Equation 4, we can again utilize Phragmén-Lindelöf to get

$$\zeta(\sigma + it) \ll |t|^{\frac{1-\sigma}{2}} \qquad 0 < \sigma < 1 \tag{11}$$

which in particular means

$$\zeta(\tfrac{1}{2} + it) \ll_\epsilon |t|^{\frac{1}{4}} \tag{12}$$

The Dedekind zeta function

In this section, we will generalize the Riemann zeta function to quadratic number fields $\mathbb{Q}(\sqrt{D})$. We will assume throughout that D is square-free and $[\mathbb{Q}(\sqrt{D}) : \mathbb{Q}] = n$. It turns out, every quadratic extension of the rationals possesses a subring that generalizes the integers.

Definition. *The* ***set of quadratic integers*** $\mathcal{O}_K$ *contained in the field* $K = \mathbb{Q}(\sqrt{D})$ *is defined to be*

$$\mathcal{O}_K := \left\{\alpha \in K \mid \alpha^2 + B\alpha + C = 0, B, C \in \mathbb{Z}\right\}$$

Observe that the quadratic formula implies

$$\alpha = \tfrac{-B\pm\sqrt{B^2-4C}}{2}$$

Proposition 3. *For $K = \mathbb{Q}[\sqrt{D}]$, D square free, there are two possible cases for the ring of integers $\mathcal{O}_K$:*

(i) if $D \equiv 2, 3 \,(mod\ 4)$, then $\mathcal{O}_K = \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$

(ii) if $D \equiv 1 \,(mod\ 4)$, then $\mathcal{O}_K = \{\frac{a+b\sqrt{D}}{2} \mid a, b \in \mathbb{Z}, a \equiv b \,(mod\ 2)\}$

Proof. We will, for the sake of computational convenience, declare $B^2 - 4C = DE^2$ with $E \in \mathbb{Z}$, meaning our roots α are of form

$$\alpha = \frac{-B \pm E\sqrt{D}}{2}$$

The quadratic formula also implies that

$$C = \frac{-B + E\sqrt{D}}{2} \cdot \frac{-B - E\sqrt{D}}{2} = \frac{B^2 - E^2 D}{4}.$$

We recall that D square-free by assumption and proceed with the cases.

(i) $D \equiv 2, 3 \,(\text{mod } 4)$

This tells us that either $B^2 \equiv 2E^2 \,(\text{mod } 4)$ or $B^2 \equiv 3E^2 \,(\text{mod } 4)$. Since neither 2 nor 3 are congruent to squares mod 4, we know that $B^2 \equiv E^2 \equiv 2 \,(\text{mod } 4)$, so B and E are even, meaning

$$\frac{-B + E\sqrt{D}}{2} = a + b\sqrt{D} \qquad a, b \in \mathbb{Z}$$

(ii) $D \equiv 1 \,(\text{mod } 4)$

This means that $B^2 \equiv E^2 \,(\text{mod } 4)$, which is equivalent to $B \equiv E \,(\text{mod } 2)$. Relabeling our variables yields the desired result.

■

Definition. *A **Dedekind domain** is a ring R satisfying the following conditions:*

(i) R is Noetherian

(ii) Every non-trivial prime ideal of R is maximal

(iii) R is integrally closed in its field of fractions (i.e. each element is the root of a monic with coefficients in its field of fractions)

Theorem 13. *The ring $\mathcal{O}_K$ is a Dedekind domain.*

Proof. There is a proof given in some lecture notes written by Brian Osserman, which may be found here. ■

Definition. *Letting K be a number field with ring of integers $\mathcal{O}_K$, the* ***absolute norm of an ideal*** *$\mathfrak{I} \subset \mathcal{O}_K$, denoted $N(\mathfrak{I})$, is defined to be*

$$N(a) := [\mathcal{O}_L : \mathfrak{I}] = |\mathcal{O}_K/\mathfrak{I}|$$

We should observe that the fact this function does not take on infinite values follows from a corollary to the linked proof that $\mathcal{O}_K$ is a Dedekind domain.

Our discussion of Dedekind domains is relevant because we can express the Riemann zeta function as a product of the norms of prime ideals.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \sum_{\substack{(n)\subset\mathbb{Z} \\ (n)\neq 0}} \frac{1}{(N(n))^s} = \prod_{\mathfrak{p} \text{ prime}} \Big(1 - \frac{1}{(N\mathfrak{p})^s}\Big)^{-1} = \prod_{n=0}^{\infty} \Big(1 - \frac{1}{p^s}\Big)^{-1}$$

It's worth going into some detail about why we can express the zeta function in this manner. We first demonstrate that

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

The proof falls out from the observation that

$$\zeta(s) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \dots$$

implies that

$$\frac{1}{2^s}\zeta(s) = \frac{1}{2^s} + \frac{1}{4^s} + \frac{1}{6^s} + \dots$$

which means

$$\Big(1 - \frac{1}{2^s}\Big)\zeta(s) = 1 + \frac{1}{3^s} + \frac{1}{5^s} + \dots$$

Clearly by repeating this process for every prime, we get

$$\prod_{p \text{ prime}} \Big(1 - \frac{1}{p^s}\Big)\zeta(s) = 1 \qquad \Rightarrow \qquad \zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Now given some $(n) \subset \mathbb{Z}$, we know that $N(n) = |\mathbb{Z}/(n)| = n$. We already know from basic abstract algebra that (n) may be expressed as a product of ideals $(p_1), \dots, (p_n)$ where $p_i | n$ is a prime integer, but in the general case we make use of the fact that every non-trivial ideal I of a Dedekind domain may be uniquely expressed as a product of prime ideals.[1] Since the norms of ideals of a Dedekind domain are multiplicative, i.e. $N\mathfrak{a} \cdot N\mathfrak{b} = N\mathfrak{a}\mathfrak{b}$, it follows that

$$N(n) = \prod_{p|n} N(p).$$

[1] In some sources, this is given as the definition of a Dedekind domain, but it can be proven as a theorem from the definition provided.

We earlier showed that the Dirichlet series equals the Euler product, which allows us to conclude

$$\zeta(s) = \prod_{(p)\subset\mathbb{Z}} \Big(1 - \frac{1}{(N(p))^s}\Big)^{-1},$$

so the zeta function has a natural characterization in the language of prime ideals.

This characterization is how we extend the zeta function to arbitrary number fields.

Definition. *Given an algebraic number field K with ring of integers $\mathcal{O}_K$, its* ***Dedekind zeta function****, $\zeta_K(s)$, is defined for complex numbers $Re(s) > 1$ by the Dirichlet series*

$$\zeta_K(s) := \sum_{0\neq\mathfrak{a}\subseteq\mathcal{O}_k} \frac{1}{(N\mathfrak{a})^s} = \prod_{\mathfrak{p}} \Big(1 - \frac{1}{(N\mathfrak{p})^s}\Big)^{-1} = \prod_{p}\prod_{\mathfrak{p}|(p)} \Big(1 - \frac{1}{(N\mathfrak{p})^s}\Big)^{-1}$$

Proposition 4. *$\zeta_K(s)$ is well-defined.*

Proof. In order for this function to be well-defined, we need its defining series to converge absolutely for $Re(s) > 1$. We're going to bound $\zeta_K(s)$ by determining the minimal value of the denominator. Recalling $|\mathbb{Q}(\sqrt{D}) : \mathbb{Q}| = n$ we observe that for some fixed $\mathfrak{p}$, $N\mathfrak{p}$ is smallest when (p) splits completely, i.e. $(p) = \prod_{i=1}^{n} \mathfrak{p}^{(i)}$ and $N\mathfrak{p} = p$. Then since all n powers of $\mathfrak{p}$ divide (p) and each power of $\mathfrak{p}$ is a prime ideal, we obtain an upper bound

$$\sum_{\mathfrak{p}} \frac{1}{(N\mathfrak{p})^s} \leqslant n \sum_{p} \frac{1}{p^s},$$

which means that the sum converges absolutely by absolute convergence of $\sum_p p^{-s}$ on $Re(s) > 1$. ■

Definition. *We say a number q is a* ***quadratic residue mod*** $\mathbf{p}$ *if there is an integer x such that $x^2 \equiv q \ (mod\ p)$. The* ***quadratic residue symbol*** *is a function*

$$\left(\tfrac{D}{p}\right) = \begin{cases} 1 & \text{if } D \text{ is a quadratic residue modulo } p \\ -1 & \text{if } D \text{ is a non-quadratic residue modulo } p \\ 0 & \text{if } D \equiv 0 \ (mod\ p) \end{cases}$$

For every rational prime $p \in \mathbb{Z}$, there are three possibilities for the corresponding prime ideal:

1. If (p) doesn't factor in $K = \mathbb{Q}[\sqrt{D}]$, then $N\mathfrak{p} = p^2$ and

$$\Big(1 - \frac{1}{(N\mathfrak{p})^s}\Big)^{-1} = \Big(1 - \frac{1}{p^{2s}}\Big)^{-1}.$$

This happens when $p \nmid D$ and the quadratic residue symbol, $\left(\frac{D}{p}\right)$, is equal to -1.

2. If $(p) = \mathfrak{p}\overline{\mathfrak{p}}$ then $N\mathfrak{p} = p$ and $\left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1} = \left(1 - \frac{1}{p^s}\right)^{-2}$.

   This happens if $p \nmid D$ and $\left(\frac{D}{p}\right) = 1$.

3. If $(p) = \mathfrak{p}^2$ then $N\mathfrak{p} = p$ and $\left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1} = \left(1 - \frac{1}{p^s}\right)^{-1}$.

   This happens if $p|D$.

The categorization we've offered above allows us to craft the definition below.

Definition. *The* ***Quadratic Character*** *associated to* $K = \mathbb{Q}(\sqrt{D})$ *and denoted* χ_D *is defined to be*

$$\chi_D(p) = \begin{cases} 1 & \text{if } (p) = \mathfrak{p}\overline{\mathfrak{p}} \\ -1 & \text{if } (p) = (p) \\ 0 & \text{if } (p) = \mathfrak{p}^2 \end{cases}$$

Here we're merely associating the ideal (p) *with the value of its associated quadratic symbol.*

It turns out, in each of the cases above, we can express $\left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1}$ in terms of the quadratic character of $\mathbb{Q}[\sqrt{D}]$.

1. If $(p) = (p)$ then $N\mathfrak{p} = p^2$ and

$$\left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1} = \left(1 - \frac{1}{p^{2s}}\right)^{-1} = \left(1 - \frac{1}{p^s}\right)^{-1}\left(1 - \frac{\chi_D(p)}{p^s}\right)^{-1}$$

2. If $(p) = \mathfrak{p}\overline{\mathfrak{p}}$ then $N\mathfrak{p} = p$ and

$$\left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1} = \left(1 - \frac{1}{p^s}\right)^{-2} = \left(1 - \frac{1}{p^s}\right)^{-1}\left(1 - \frac{\chi_D(p)}{p^s}\right)^{-1}$$

3. If $(p) = \mathfrak{p}^2$ then $N\mathfrak{p} = p$ and

$$\left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1} = \left(1 - \frac{1}{p^s}\right)^{-1} = \left(1 - \frac{1}{p^s}\right)^{-1}\left(1 - \frac{\chi_D(p)}{p^s}\right)^{-1}$$

This means that

$$\zeta_K(s) = \left(\prod_{\substack{p \text{ prime} \\ (p)=\mathfrak{p}}} \left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1}\right)\left(\prod_{\substack{p \text{ prime} \\ (p)=\mathfrak{p}\overline{\mathfrak{p}}}} \left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1}\right)\left(\prod_{\substack{p \text{ prime} \\ (p)=\mathfrak{p}^2}} \left(1 - \frac{1}{(N\mathfrak{p})^s}\right)^{-1}\right)$$

$$= \prod_p \left(1 - \frac{1}{p^s}\right)^{-1} \prod_p \left(1 - \frac{\chi_D(p)}{p^s}\right)^{-1}$$

Observe the presence of the product form of the ordinary Riemann zeta function as a factor, yielding the characterization

$$\zeta_K(s) = \zeta(s) \prod_p \left(1 - \frac{\chi_D(p)}{p^s}\right)^{-1} = \zeta(s) L(s, \chi_D) \tag{14}$$

Where

$$L(s, \chi_D) := \sum_{n=1}^{\infty} \frac{\chi_D(n)}{n^s}$$

is called the **L-series associated with $\chi_{\mathbf{D}}$**. The equality in Equation 14 may be seen from observing that

$$\left(1 - \tfrac{\chi_D(p)}{p^s}\right)^{-1} = \sum_{n=0}^{\infty} \frac{\chi_D(p)}{p^{ns}}$$

which means, in a similar application of Euler product magic to that seen earlier:

$$\prod_p \left(1 - \frac{\chi_D(p)}{p^s}\right)^{-1} = \prod_p \left(\sum_{n=0}^{\infty} \frac{\chi_D(p)}{p^{ns}}\right) = \sum_{n=1}^{\infty} \frac{\chi_D(p)}{n^s}$$

The Functional Equation of ζ_K

Just as in the case of the standard Riemann-zeta function, we can define the analytic continuation of ζ_K to complex numbers with real part less than 1 by deploying the functional equation of ζ_K, defined by the equation

$$\xi_K(s) = \begin{cases} s(1-s)\left(\frac{|D|}{(2\pi)^2}\right)^{\frac{s}{2}} \Gamma(\frac{s}{2}) \zeta_K(s) & D < 0 \\ \\ s(1-s)\left(\frac{|D|}{\pi^2}\right)^{\frac{s}{2}} \Gamma(\frac{s}{2}) \zeta_k(s) & D > 0 \end{cases}$$

Theorem 15. *Over $K = \mathbb{Q}(\sqrt{D})$, $\xi_K(s) = \xi_K(1-s)$. That is,*

$$\left(\tfrac{|D|}{(2\pi)^2}\right)^{\frac{s}{2}} \Gamma(\tfrac{s}{2}) \zeta_K(s) = \left(\tfrac{|D|}{(2\pi)^2}\right)^{\frac{1-s}{2}} \Gamma(\tfrac{1-s}{2}) \zeta_K(1-s) \qquad \textit{if } D < 0$$

$$\left(\tfrac{|D|}{\pi^2}\right)^{\frac{s}{2}} \Gamma(\tfrac{s}{2}) \zeta_K(s) = \left(\tfrac{|D|}{\pi^2}\right)^{\frac{1-s}{2}} \Gamma(\tfrac{1-s}{2}) \zeta_K(1-s) \qquad \textit{if } D > 0$$

Proof. The proof basically falls out from Equation 14 and the fact that the ordinary Riemann zeta function satisfies a similar property. ■

For reasons to be seen later, we remark that

$$\operatorname*{Res}_{s=1} \zeta_K(s) = L(1, \chi_D) = \frac{2\pi h_k}{\omega\sqrt{D_K}} \tag{16}$$

where h_K is the class number of K, ω is the number of roots of 1 in K, and

$$D_K = \begin{cases} |D| & D \equiv 1 \pmod 4 \\ 4|D| & D \equiv 2, 3 \pmod 4 \end{cases} \tag{17}$$

Bounding $\zeta_K(s)$ near the the critical strip

We consider the case where $D < 0$ and fix $\epsilon > 0$ and set $s = -\epsilon + it$.[2] We now make use of Theorem 15 to make a nearly identical argument to that we made earlier in Equation 6, giving us:

$$\zeta_K(-\epsilon + it) = \left(\frac{|D|}{(2\pi)^2}\right)^{1+\epsilon-it} \frac{\Gamma(\frac{1+\epsilon-it}{2})}{\Gamma(\frac{-\epsilon+it}{2})} \zeta_K(1 + \epsilon - it) \ll_\epsilon |D|^{1+\epsilon} \left|\frac{\Gamma(\frac{1+\epsilon-it}{2})}{\Gamma(\frac{-\epsilon+it}{2})}\right|$$

We can get an even better bound by using Stirling's Formula again to conclude that

$$\left|\frac{\Gamma(\frac{1+\epsilon-it}{2})}{\Gamma(\frac{-\epsilon+it}{2})}\right| \sim \left|\frac{t}{2}\right|^{1+2\epsilon} \ll_\epsilon |t|^{1+2\epsilon} \tag{18}$$

Since our ϵ is some arbitrarily small positive number, we obtain following bound of $\zeta_K(s)$ for values s that are "immediately to the left" of the critical strip and have large imaginary component, that is, as t gets large, we have the bound

$$\boxed{\zeta_K(-\epsilon + it) \ll_\epsilon |D|^{1+\epsilon}|t|^{1+2\epsilon} \ll_\epsilon |D|^{\frac{1}{2}+2\epsilon}|t|^{1+2\epsilon}} \tag{19}$$

For the other side of the critical strip, we again deploy a symmetry argument to get

$$\zeta_K(1 - s) = \frac{\left(\frac{|D|}{(2\pi)^2}\right)^{\frac{s}{2}} \Gamma(\frac{s}{2})}{\left(\frac{D}{(2\pi)^2}\right)^{\frac{1-s}{2}} \Gamma(\frac{1-s}{2})} \zeta_K(s) = \left(\frac{|D|}{(2\pi)^2}\right)^{s-\frac{1}{2}} \frac{\Gamma(\frac{s}{2})}{\Gamma(\frac{1-s}{2})} \zeta_K(s)$$

Equation 18 implies

$$\frac{\Gamma(\frac{-\epsilon+it}{2})}{\Gamma(\frac{1+\epsilon-it}{2})} \ll_\epsilon \frac{1}{|t|^{1+2\epsilon}}.$$

Together with the bound on $\zeta_K(-\epsilon + it)$ seen in Equation 19, we have a new bound

[2] The logic is equivalent when $D > 0$.

$$\zeta_K(1+\epsilon-it) \ll_\epsilon \left(\frac{|D|}{(2\pi)^2}\right)^{\frac{1}{2}+\epsilon-it} \frac{1}{|t|^{1+2\epsilon}} \zeta_K(-\epsilon+it) \ll_\epsilon \left(\frac{|D|}{(2\pi)^2}\right)^{\frac{1}{2}+\epsilon-it} \frac{|D|^{\frac{1}{2}+\epsilon}|t|^{1+2\epsilon}}{|t|^{1+2\epsilon}} = \left(\frac{|D|}{(2\pi)^2}\right)^{1+2\epsilon-it}$$

The absolute value of this last expression is a constant which depends only on our choice of ϵ, hence we have that

$$\zeta_K(1+\epsilon-it) \ll_\epsilon 1 = |D|^0|t|^0 \tag{20}$$

At this point, we have bounds for ζ_K on either side of the critical strip. We can use these to glean a bit of information about the behavior on the interior of the critical strip using the Phragmén-Lindelöf principle as in the earlier discussion of ordinary Riemann zeta function. That is, Phragmén-Lindelöf yields:

$$\zeta_K(\tfrac{1}{2}+it) \ll_\epsilon |D|^{\frac{1}{4}+\epsilon}|t|^{\frac{1}{2}+\epsilon} \tag{21}$$

Bounding ζ_K within the critical strip

Choose some $x \in \mathbb{R}$ with the property $x > 1$ and suppose we have a β such that $\zeta_K(\beta) \leqslant 0$. We will use it to construct the contour integral along the vertical line consisting of complex numbers with $\mathrm{Re}(s) = 2$:

$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} = \frac{1}{2\pi i}\int\limits_{(2)} \frac{\sum\limits_{n\geqslant 1}\frac{a_n}{n^{s+\beta}}x^s ds}{s(s+1)(s+2)}$$

It follow from this that

$$\Big|\sum_{n\geqslant 1}\frac{a_n}{n^{s+\beta}}\Big| = |\zeta_K(s+\beta)|$$

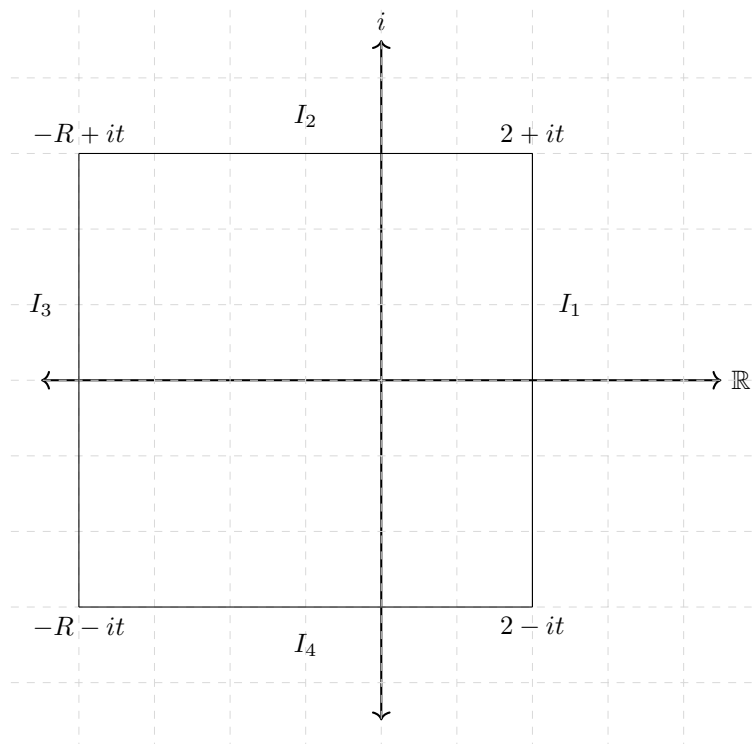
Therefore, we can change the order of summation and integration, so

$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{\sum\limits_{n\geqslant 1}\frac{a_n}{n^{s+\beta}}x^s ds}{s(s+1)(s+2)} = \frac{1}{2\pi i}\left(\sum_{n\geqslant 1}\frac{a_n}{n^\beta}\int\limits_{(2)} \frac{\left(\frac{x}{n}\right)^s ds}{s(s+1)(s+2)}\right)$$

Given some $x \geqslant 0$ we have

$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{x^s ds}{s(s+1)(s+2)} = \begin{cases} \frac{1}{2}\left(1-\frac{1}{y}\right)^2 & y \geqslant 1 \\ 0 & 0 \leqslant y < 1 \end{cases}$$

We can make a quick, rigorous argument demonstrating this. Consider the rectangular boundary curve $C(t,r)$ pictured below:



$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{x^s ds}{s(s+1)(s+2)} \text{ has poles at } s=0,1, \text{ and } 2$$

The residues at these poles are as follows:

$$\operatorname*{Res}_{s=0}\frac{x^s}{s(s+1)(s+2)} = \lim_{s\to 0}\frac{x^s}{(s+1)(s+2)} = \frac{1}{2}$$
$$\operatorname*{Res}_{s=-1}\frac{x^s}{s(s+1)(s+2)} = \lim_{s\to -1}\frac{x^s}{s(s+2)} = -\frac{1}{x}$$
$$\operatorname*{Res}_{s=-2}\frac{x^s}{s(s+1)(s+2)} = \lim_{s\to -2}\frac{x^s}{s(s+1)} = \frac{1}{2x^2}$$

The Cauchy Residue theorem then tells us that

$$\frac{1}{2\pi i}\int\limits_{C(t,r)} \frac{x^s}{s(s+1)(s+2)} = \frac{1}{2}-\frac{1}{x}+\frac{1}{x^2} = \frac{1}{2}\Big(1-\frac{1}{x}\Big)^2$$

Taking a look again at

$$\frac{1}{2\pi i}\left(\sum_{n\geqslant 1}\frac{a_n}{n^\beta}\int\limits_{(2)}\frac{\left(\frac{x}{n}\right)^s ds}{s(s+1)(s+2)}\right),$$

we note that the summands in the numerator of the latter integral will approach 0 after $n > x$. With this and the fact that $a_1 = 1$ and $1^s = 1$, we can rewrite the nasty sum of integrals as

$$\frac{1}{2}\sum_{n<x}\frac{a_n}{n^\beta}\Big(1-\frac{n}{x}\Big)^2 > \frac{1}{2}\Big(1-\frac{1}{x}\Big)^2$$

Which gives us

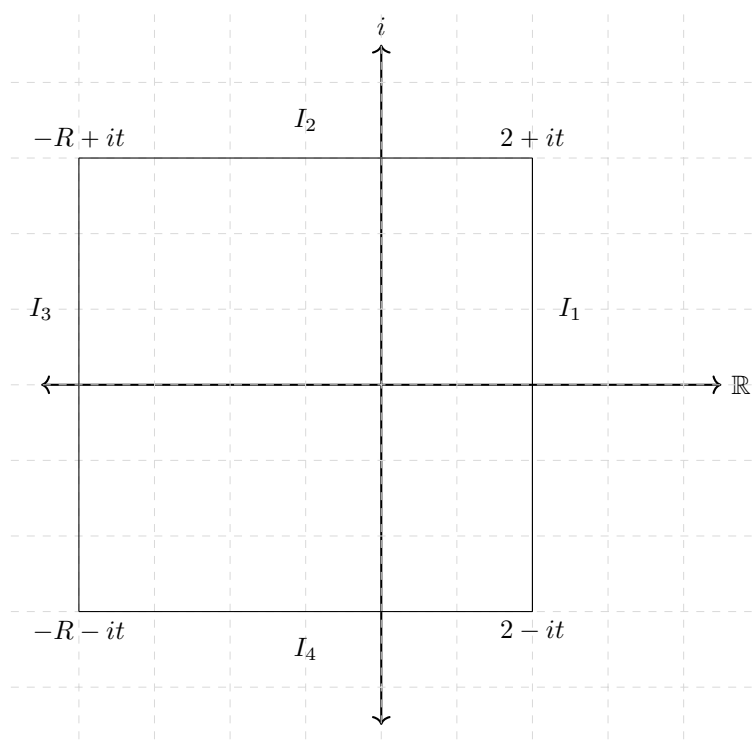
$$\lim_{x\to\infty}\frac{1}{2}\sum_{n<x}\frac{a_n}{n^\beta}\Big(1-\frac{n}{x}\Big)^2 > \lim_{x\to\infty}\frac{1}{2}\Big(1-\frac{1}{x}\Big)^2 = \frac{1}{2}$$

Thanks to the functional equation for the zeta function, by Phragmen-Linelof, we have

$$\Big|\zeta_K(\frac{1}{2}+it)\Big| \ll |D|^{\frac{1}{4}+\epsilon}|t|^{\frac{1}{2}+\epsilon} \qquad \text{as } |t|\to\infty \tag{22}$$

$$\frac{1}{2\pi i}\int\limits_{(2)}\frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} = \begin{cases}\frac{1}{(\mathrm{Re}(s))!}\Big(1-\frac{1}{x}\Big) & x \geqslant 1\\ 0 & 0 < x \leqslant 1\end{cases}$$

We consider the rectangular boundary curve $C(t,r)$ pictured below:



$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} \text{ has poles at } s = 0, 1, \text{ and } 2$$

we compute the residues at these poles as follows:

$$\operatorname*{Res}_{s=0} \frac{x^s}{s(s+1)(s+2)} = \lim_{s\to 0} \frac{x^s}{(s+1)(s+2)} = \frac{1}{2}$$

$$\operatorname*{Res}_{s=-1} \frac{x^s}{s(s+1)(s+2)} = \lim_{s\to -1} \frac{x^s}{s(s+2)} = -\frac{1}{x}$$

$$\operatorname*{Res}_{s=-2} \frac{x^s}{s(s+1)(s+2)} = \lim_{s\to -2} \frac{x^s}{s(s+1)} = \frac{1}{2x^2}$$

The definition of a contour integral tells us that

$$\frac{1}{2\pi i}\int_{C(t,r)} \frac{x^s}{s(s+1)(s+2)} = I_1 + I_2 + I_3 + I_4$$

Now, letting $s = \sigma + iT$ and $ds = d\sigma$, we have

$$|I_2| = \left|\frac{1}{2\pi i}\int\limits_{\sigma=2}^{-R} \frac{x^{\sigma+iT}d\sigma}{(\sigma+iT)(\sigma+1+iT)(\sigma+2+iT)}\right| \ll x^2 \int\limits_{\sigma=2}^{-R} \frac{d\sigma}{(\sigma+iT)(\sigma+1+iT)(\sigma+2+iTv)}$$

and since

$$x^2 \int\limits_{\sigma=2}^{-R} \frac{d\sigma}{(\sigma+iT)(\sigma+1+iT)(\sigma+2+iTv)} \ll \frac{x^2 T}{T^3} = \frac{x^2}{T^2}$$

which approaches 0 as T gets huge. We note that $|I_4|$ will evaluate to exactly the same value.

Now letting $s = -T + it$ and $ds = idt$, we have

$$|I_3| = \left|\frac{1}{2\pi i}\int\limits_{t=-T}^{T} \frac{x^{-T+it}dt}{(-T+it)(-T+1+it)(-T+2+it)}\right| \ll x^{-T}\frac{T}{T^3}$$

which also approaches 0 as T gets huge.

From these investigations, we conclude that

$$\frac{1}{2\pi i}\int\limits_{C(r,t)} \frac{x^s ds}{s(s+1)(s+2)} = I_1 + I_2 + I_3 + I_4 = I_1 + 0 + 0 + 0$$

Then the Cauchy residue theorem tells us that

$$\frac{1}{2\pi i}\int\limits_{C(r,t)} \frac{x^s ds}{s(s+1)(s+2)} = I_1 = \sum \text{Residues}$$

So now, if $0 \leqslant x < 1$, then we quickly see that the x^s in the numerator drives everything to 0.

So we have

$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} > \frac{1}{4}$$

Moving the line of integration from (2) to $(\frac{1}{2} - \beta)$

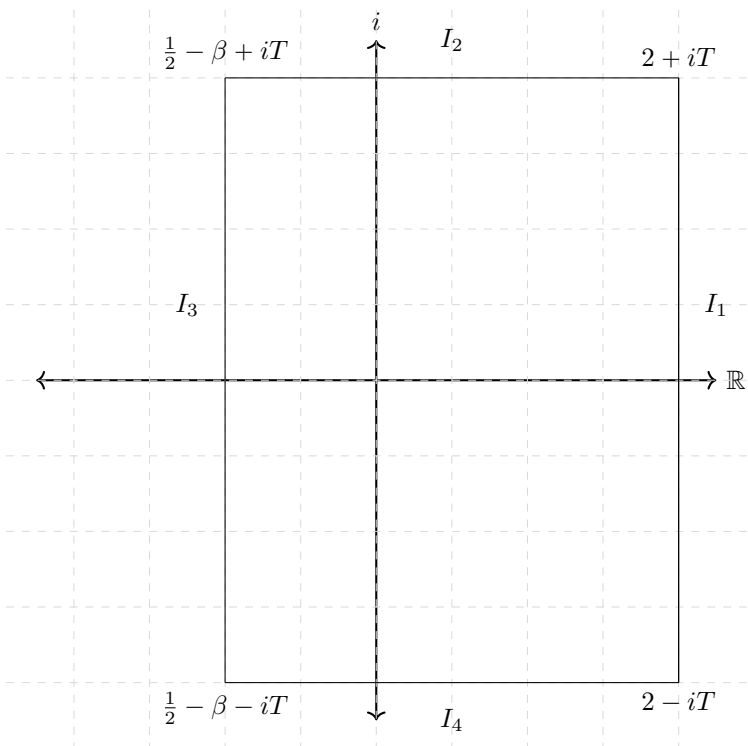
We already know by some previous work that

$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{x^s}{s(s+1)(s+2)} = \begin{cases} \frac{1}{2}(1-\frac{1}{x})^2 & , x \geqslant 1 \\ 0 & , 0 < x \leqslant 1 \end{cases}$$

As it turns out, this generalizes:

$$\frac{1}{2\pi i}\int\limits_{(2)} \frac{x^s}{s(s+1)\dots(s+r)} = \begin{cases} \frac{1}{r!}(1-\frac{1}{x})^r & , x \geqslant 1 \\ 0 & , 0 < x \leqslant 1 \end{cases}$$

We're now going to consider the integral around the contour given below.



Since the sum of the residues is equal to the contour integral, and since I_2 and I_4 tend to zero by a similar argument to that given earlier, we have

$$I_1 + I_3 = \sum \text{Residues}$$

We now need to determine whether there are any poles enclosed by the given contour integral. We know that there is a pole at $s = 0$ and it turns out that

$$\operatorname*{Res}_{s=0} \frac{\zeta_K(s+\beta)}{s(s+1)(s+2)(s+2)} = \frac{\zeta_K(\beta)}{2}.$$

There is also a residue at $s = 1 - \beta$ which we compute to be:

$$\operatorname*{Res}_{s=1-\beta} \frac{\zeta_K(s+\beta)x^s}{s(s+1)(s+2)} = \operatorname*{Res}_{s=1} \frac{\zeta_K(s)x^{1-\beta}}{(1-\beta)(2-\beta)(3-\beta)} = \frac{L(1,\chi_{-D})x^s}{(1-\beta)(2-\beta)(3-\beta)}$$

Hence we have

$$\frac{1}{4} < \frac{1}{2\pi i} \int\limits_{(2)} \frac{\zeta_K(s+\beta)x^s}{s(s+1)(s+2)} = \frac{1}{2\pi i} \int\limits_{(\frac{1}{2}-\beta)} \frac{\zeta_K(s+\beta)x^s}{s(s+1)(s+2)} + \frac{L(1,\chi_{-D})x^s}{(1-\beta)(2-\beta)(3-\beta)} - \frac{\zeta_K(\beta)}{2}$$

We've now moved the line of integration to $Re(s+\beta) = \frac{1}{2}$ or in other words $Re(s) = \frac{1}{2} - \beta$. We note that $\frac{1}{2} - \beta$ is basically $-\frac{1}{2} + \epsilon$, where ϵ is some tiny positive real.

$$\mathcal{I} = \int\limits_{Re(s+\beta)=\frac{1}{2}} \frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} + \frac{\zeta_K(\beta)}{2} + \frac{\left(\underset{s=1-\beta}{Res}\,\zeta_K(s+\beta)\right)x^{1-\beta}}{(1-\beta)(2-\beta)(3-\beta)}$$

And since $\underset{s=1-\beta}{\text{Res}}\,\zeta_K(s+\beta) = L(1,\chi_{-D})$, we conclude that

$$\mathcal{I} = \int\limits_{Re(s+\beta)=\frac{1}{2}} \frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} + \frac{\zeta_K(\beta)}{2} + \frac{L(1,\chi_{-D})x^{1-\beta}}{(1-\beta)(2-\beta)(3-\beta)}$$

Now we need to determine how big $\zeta_K(s+\beta)$ is. We know that $Re(s+\beta) = \frac{1}{2}$, so $s+\beta = \frac{1}{2} + it$ where t is allowed to vary.
$s = -\frac{1}{2} + \epsilon + it$ where $\epsilon = 1-\beta$, allowing us to write

$$\int\limits_{Re(s+\beta)=\frac{1}{2}} \frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} = \int\limits_{-\infty}^{\infty} \frac{\zeta_K(\frac{1}{2}+it)x^{-\frac{1}{2}+\epsilon+it}dt}{(\frac{1}{2}-\beta+it)(\frac{3}{2}-\beta+it)(\frac{5}{2}-\beta+it)}$$

We recall the convexity bound given by Equation 22:

$$\zeta_K(\frac{1}{2}+it) \ll t^{\frac{1}{4}+\epsilon}D^{\frac{1}{2}+\epsilon}$$

$$\int\limits_{-\infty}^{\infty} \frac{\zeta_K(\frac{1}{2}+it)x^{-\frac{1}{2}+\epsilon+it}dt}{(\frac{1}{2}-\beta+it)(\frac{3}{2}-\beta+it)(\frac{5}{2}-\beta+it)} \ll \int\limits_{-\infty}^{\infty} \frac{t^{\frac{1}{4}+\epsilon}D^{\frac{1}{4}+\epsilon}x^{-\frac{1}{2}+\epsilon}dt}{(\frac{1}{2}-\beta+it)(\frac{3}{2}-\beta+it)(\frac{5}{2}-\beta+it)}.$$

Since

$$\frac{1}{4} < \frac{1}{2\pi i}\int\limits_{(2)} \frac{\zeta_K(s+\beta)x^s ds}{s(s+1)(s+2)} = \mathcal{I}$$

and since $\frac{\zeta_K(\beta)}{2}$ is a constant, as is $\frac{L(1,\chi_{-D})x^{1-\beta}}{(1-\beta)(2-\beta)(3-\beta)}$ due to the fact that $x = D^a$, we therefore have

$$\frac{1}{4} < \mathcal{I} \ll \left(\int\limits_{-\infty}^{\infty} \frac{|t|^{\frac{1}{4}+\epsilon}|D|^{\frac{1}{4}+\epsilon}x^{-\frac{1}{2}+\epsilon}dt}{\left(\frac{1}{2}-\beta+it\right)\left(\frac{3}{2}-\beta+it\right)\left(\frac{5}{2}-\beta+it\right)}\right) + \frac{\zeta_K(\beta)}{2} + \frac{L(1,\chi_{-D})x^{1-\beta}}{(1-\beta)(2-\beta)(3-\beta)}$$

Finally, we know that the integral

$$\int\limits_{-\infty}^{\infty} \frac{t^{\frac{1}{4}+\epsilon}dt}{\left(\frac{1}{2}-\beta+it\right)\left(\frac{3}{2}-\beta+it\right)\left(\frac{5}{2}-\beta+it\right)}$$

converges because each component of the denominator stays away from zero and the degree of the denominator exceeds that of the numerator. Hence, we

substitute $x = D^a$ to arrive at

$$\int_{-\infty}^{\infty} \frac{t^{\frac{1}{4}+\epsilon} D^{\frac{1}{4}+\epsilon} x^{-\frac{1}{2}+\epsilon} dt}{\left(\frac{1}{2}-\beta+it\right)\left(\frac{3}{2}-\beta+it\right)\left(\frac{5}{2}-\beta+it\right)} = \int_{-\infty}^{\infty} \frac{t^{\frac{1}{4}+\epsilon} D^{\frac{1}{4}+\epsilon} D^{-\frac{a}{2}+a\epsilon} dt}{\left(\frac{1}{2}-\beta+it\right)\left(\frac{3}{2}-\beta+it\right)\left(\frac{5}{2}-\beta+it\right)} \ll D^{\frac{1}{4}+\epsilon} D^{-\frac{a}{2}+a\epsilon}$$

so we can conclude that

$$\frac{1}{4} < \mathcal{I} \ll D^{\frac{1}{4}+\epsilon} D^{-\frac{a}{2}+a\epsilon} + \frac{\zeta_K(\beta)}{2} + \frac{L(1,\chi_{-D}) x^{1-\beta}}{(1-\beta)(2-\beta)(3-\beta)}$$

We would like $D^{\frac{1}{4}+\epsilon} D^{-\frac{a}{2}+a\epsilon}$ to become small. Since $\epsilon = 1-\beta$ is going to be very, very small, it basically contributes nothing. So basically, we want $\frac{-a}{2} + \frac{1}{4} < 0$.

Example 1. *If I let $a = 3$, then $D^{\frac{1}{4}+\epsilon} D^{-\frac{a}{2}+a\epsilon} = D^{-\frac{3}{2}+\frac{1}{4}+4\epsilon}$*

It follows that

$$D^{\frac{1}{4}+\epsilon} D^{-\frac{a}{2}+a\epsilon} < \frac{1}{8}$$

which in turn means that

$$\frac{1}{4} < \frac{1}{8} + \frac{\zeta_K(\beta)}{2} + \frac{L(1,\chi_{-D}) x^{1-\beta}}{(1-\beta)(2-\beta)(3-\beta)}$$

from which it follows that

$$\frac{L(1,\chi_{-D}) D^{a(1-\beta)}}{(1-\beta)(2-\beta)(3-\beta)} > \frac{1}{8} - \frac{\zeta_K(\beta)}{2}.$$

We chose β such that $\zeta_K(\beta) \leqslant 0$, so we know that no matter what, we have

$$\frac{L(1,\chi_{-D}) D^{a(1-\beta)}}{(1-\beta)(2-\beta)(3-\beta)} > \frac{1}{8} - \frac{\zeta_K(\beta)}{2} \geqslant \frac{1}{8}$$

Using the fact that

$$D^{a(1-\beta)} = e^{log(D^{a(1-\beta)})} = e^{a(1-\beta)log(D)},$$

we can write

$$\frac{L(1,\chi_{-D}) D^{a(1-\beta)}}{(1-\beta)(2-\beta)(3-\beta)} = \frac{L(1,\chi_{-D})}{1-\beta} \cdot \frac{e^{a(1-\beta)log(D)}}{(2-\beta)(3-\beta)}$$

Since $2-\beta \approx 1$ and $3-\beta \approx 2$, we can ignore them. Recalling **??**, that is,

$$1 - \frac{1}{clog|D|} < \beta < 1$$

from which we derive

$$1 - \beta < \frac{1}{clog|D|}$$

This means that $log|D|(1-\beta)a < \frac{a}{c}$ which in turn means $e^{a(1-\beta)log|D|} < e^{\frac{a}{c}}$, so

$$\frac{1}{8} < \frac{L(1,\chi_{-D})}{1-\beta} \cdot \frac{e^{a(1-\beta)log(D)}}{(2-\beta)(3-\beta)} \ll \frac{L(1,\chi_{-D})}{1-\beta}$$

This is equivalently expressed as

$$\frac{L(1,\chi_{-D})}{1-\beta} \gg 1,$$

which in turn means that

$$L(1,\chi_{-D}) \gg 1-\beta.$$

Recall Equation 16:

$$L(1,\chi_D) = \frac{2\pi h_k}{\omega\sqrt{D_K}}$$

This means that

$$\frac{2\pi h_k}{\omega\sqrt{D_K}} \gg 1-\beta$$

from which follows

$$h_k \gg (1-\beta)\sqrt{D_K} \tag{23}$$

Back at the very beginning, we said that we want to prove that

$$\frac{2\pi h_K}{\omega\sqrt{D_K}} > c(1-\beta)$$

and that

$$h_K \gg_\epsilon (log|D_K|)^{1-\epsilon}.$$

We also said that we should be able to take $1-\beta = \frac{c}{log|D_K|}$ and leverage it to get $h_K > \frac{c\sqrt{|D_K|}}{log|D_K|}$.

Siegel's Theorem

Theorem 24 (Siegel's Theorem)**.** $h_K \ll_\epsilon |D_K|^{\frac{1}{2}-\epsilon}$

We're going to take a look at an argument Dorian Goldfeld offers in "A Simple Proof of Siegel's Theorem."

Let's suppose there is a D_0 such that $L(\beta,\chi_{D_0}) = 0$ for some β such that $\frac{1}{\log|D_0|} \ll \beta < 1$. We might as well assume that for some $\epsilon > 0$ that $1-\beta < \epsilon$. If $1-\beta > \epsilon$, then the lower bound would be no problem. By the way, we don't care whether D_0 is positive or negative. Now let's take $a-D$ where a and D are squarefree and $-D < 0$. Then we want to take a look at three quadratic fields: $\mathbb{Q}[\sqrt{-D}]$, $\mathbb{Q}[\sqrt{D_0}]$, and their composite field, $\mathbb{Q}[\sqrt{-D},\sqrt{D_0}]$. (There's another field sitting inside of the final one, it is $\mathbb{Q}[\sqrt{-DD_0}]$).

Let $K = \mathbb{Q}[\sqrt{-D}]$, $K_0 = \mathbb{Q}[\sqrt{D_0}]$, $L = \mathbb{Q}[\sqrt{-D}, \sqrt{D_0}]$, and $K' = \mathbb{Q}[\sqrt{-DD_0}]$

There is a zeta function associated to L and it is:

$$\zeta_L(s) = \sum_{0 \neq \mathfrak{a} \subset \mathcal{O}_L} \left(1 - \frac{1}{(N\mathfrak{a})^s}\right)^{-1} = \zeta(s)L(s, \chi_{-D})L(s, \chi_{D_0})L(s, \chi_{-DD_0})$$

The significance of this is that its Dirichlet series is equal to

$$\sum_{n \geqslant 1} \frac{a_n}{n^s} \qquad a_1 = 1 \text{ and } a_n \geqslant 0$$

We can write this down as

$$\zeta(s)L(s, \chi_{-D})L(s, \chi_{D_0})L(s, \chi_{-DD_0}) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1} \left(1 - \frac{\chi_{-D}(p)}{p^s}\right)^{-1} \left(1 - \frac{\chi_{D_0}(p)}{p^s}\right)^{-1} \left(1 - \frac{\chi_{-DD_0}(p)}{p^s}\right)^{-1}$$

We're going to set up an analogue to the thing that we looked at previously. Noting that in

$$\frac{1}{2\pi i} \int_{(2)} \frac{\zeta_L(s + \beta_0)x^s ds}{s(s+1)(s+2)}$$

we have

$$\zeta_L(s + \beta_0) = \sum_{n \geqslant 1} \frac{a_n}{n^{s+\beta_0}}$$

so substituting in, we find

$$\frac{1}{2\pi i} \int_{(2)} \frac{\zeta_L(s + \beta_0)x^s ds}{s(s+1)(s+2)} = \frac{1}{2\pi i} \left(\sum_{n \geqslant 1} \frac{a_n}{n^{\beta_0}} \int_{(2)} \frac{\left(\frac{x}{n}\right)^s ds}{s(s+1)(s+2)} \right) = \left(\sum_{1 \leqslant n < x} \frac{a_n}{n^{\beta_0}} \right) \frac{1}{2} \left(1 - \frac{n}{x}\right)^2 > \frac{1}{4}$$

Let's say

$$\mathcal{I} = \int_{(2)} \frac{\zeta_L(s + \beta_0)x^s ds}{s(s+1)(s+2)}$$

Just like before, we move the line of integration to $-\frac{1}{2} + \beta_0 + it$ where we have $-\infty \leqslant t \leqslant \infty$. We know that $\mathcal{I} + I_2 + I_3 + I_4$ is the sum of the residues. Where are these residues? There's a hole at $s + \beta_0 = 1$. We compute its residue by noting that

$$\zeta_L(s + \beta_0) = \zeta(s)L(s, \chi_{-D})L(s, \chi_{D_0})L(s, \chi_{-DD_0})$$

In the product, only $\zeta(s)$ has a pole at $s = 1$, and $\operatorname*{Res}_{s=1} \zeta(s) = 1$, so

$$\operatorname*{Res}_{s+\beta_0=1} \zeta_L(s+\beta_0) = L(1,\chi_{-D})L(1,\chi_{D_0})L(1,\chi_{-DD_0}) \cdot \frac{x^{1-\beta_0}}{(1-\beta_0)(2-\beta_0)(3-\beta_0)}$$

Since $I_2 = 0 = I_4$, we know that

$$\mathcal{I}+I_3 = \operatorname*{Res}_{s=1} \zeta_L(s+\beta_0) = L(1,\chi_{-D})L(1,\chi_{D_0})L(1,\chi_{-DD_0}) \cdot \frac{x^{1-\beta_0}}{(1-\beta_0)(2-\beta_0)(3-\beta_0)} := Res$$

This means that

$$\mathcal{I} = Res - I_3.$$

To bound I_3, we have to bound

$$\int_{-\infty}^{\infty} \zeta\Big(\frac{1}{2}+it\Big)L\Big(\frac{1}{2}+it,\chi_{-D}\Big)L\Big(\frac{1}{2}+it,\chi_{D_0}\Big)L\Big(\frac{1}{2}+it,\chi_{-DD_0}\Big) \cdot \frac{x^{\frac{1}{2}-\beta_0+it}}{\Big(\frac{1}{2}-\beta_0+it\Big)\Big(\frac{3}{2}-\beta+it\Big)\Big(\frac{5}{2}-\beta_0+it\Big)}$$

We now note that we have the following bounds:

1. $\zeta(\frac{1}{2}+it) \ll t^{\frac{1}{4}+\epsilon}$
2. $L(\frac{1}{2}+it,\chi_{-D}) \ll t^{\frac{1}{4}+\epsilon}D^{\frac{1}{4}+\epsilon}$
3. $L(\frac{1}{2}+it,\chi_{D_0}) \ll t^{\frac{1}{4}+\epsilon}D_0^{\frac{1}{4}+\epsilon}$
4. $L(\frac{1}{2}+it,\chi_{-DD_0}) \ll t^{\frac{1}{4}+\epsilon}(DD_0)^{\frac{1}{4}+\epsilon}$

Taken all together, we get

$$I_3 \ll \int_{-\infty}^{\infty} \frac{t^{1+4\epsilon}(DD_0)^{\frac{1}{2}+2\epsilon}x^{\frac{1}{2}-\beta+it}}{(t+1)^3}$$

Remember that β is really close to 1, so $\frac{1}{2}-\beta = -\frac{1}{2}+\epsilon'$ where ϵ' is a small positive number. So why not let $x = (DD_0)^4$. This gives us $x^{\frac{1}{2}-\beta} = (DD_0)^{-2+4\epsilon'}$, meaning

$$I_3 \ll \int_{-\infty}^{\infty} \frac{t^{1+4\epsilon}(DD_0)^{\frac{1}{2}+2\epsilon}x^{\frac{1}{2}-\beta+it}}{(t+1)^3} = \int_{-\infty}^{\infty} \frac{t^{1+4\epsilon}(DD_0)^{-\frac{3}{2}+2\epsilon+4\epsilon'}}{(t+1)^3}$$

and since the integral converges, we will have then

$$I_3 \ll D^{-\frac{3}{2}}$$

All told, we know that

$$1 \ll \mathcal{I} + I_3 = L(1,\chi_{-D})L(1,\chi_{D_0})L(1,\chi_{-DD_0}) \cdot \frac{x^{1-\beta_0}}{(1-\beta_0)(2-\beta_0)(3-\beta_0)}$$

It is not hard to show that

$$L(1, \chi_{-DD_0}) \ll log|DD_0|$$

from which it follows that $x^{1-\beta} = (DD_0)^4(1-\beta_0)$. Since D_0 and β_0 are constants, and $1-\beta_0 < \epsilon$ This gives us

$$L(1, \chi_{-D}) \gg_{D_0} (1-\beta_0)D^{4\epsilon}$$

which means that

$$h_k \gg_{D_0} (1-\beta_0)D^{4\epsilon}$$

which gives us Siegel's theorem.

The Deuring-Heilbron Repulsion Principle

A bizarre fact about Siegel zeros is that given one exists, we can show that there are no others, that they of necessity repel each other. This fact is called the Deuring-Heilbron repulsion principle or sometimes just the repulsion principle. Building up to a proof, we're going to borrow some definitions from Ahlfors' *Complex Analysis*.

Definition (From Ahlfor's *Complex Analysis*)**.** *The* ***order*** *of an entire function $f(s)$ is*

$$\lambda := \limsup_{r\to\infty} \frac{log\Big(log\big(\max_{|z|=r}|f(z)|\ \big)\Big)}{log(r)}$$

An equivalent characterization is that $\max_{|z|=r}|f(z)| = e^{r^{\lambda+\epsilon}}$ for any $\epsilon > 0$ for sufficiently large r.

Theorem 25 (Weierstrauss Factorization Theorem)**.** *Given $\{a_n\}$ is the sequence of non-zero zeros of the entire function f (repeating according to multiplicity) and letting m be the order of the zero at $z = 0$ (where $m = 0$ indicates that $f(0) \neq 0$), there is a sequence of integers $\{p_n\}$ and an entire function g such that*

$$f(z) = e^{g(z)}z^m \prod_{n=1}^{\infty}\left(1-\frac{z}{a_n}\right)exp\left(\sum_{k=1}^{p_n}\frac{z^k}{ka_n^k}\right)$$

and the product converges locally uniformly absolutely on $\mathbb{C}$.[3]

Definition. *Given an entire function f with a sequence of zeros $\{a_n\}$, the* ***genus*** *of f is the least integer h such that f may be written as*

$$f(z) = e^{g(z)}z^m \prod_{n=1}^{\infty}\left(1-\frac{z}{a_n}\right)exp\left(\sum_{k=1}^{h}\frac{z^k}{ka_n^k}\right)$$

[3] That is, for each $s \in \mathbb{C}$, there is a neighborhood on which the product converges uniformly and absolutely.

with $g(z) \in \mathbb{C}[z]$ and $deg(g) \leqslant h$. If no such reformulation of f exists, we say that f has infinite genus.

It turns out that the order of an entire function is more intimately related to the genus than is immediately clear.

Theorem 26 (Hadamard's Theorem)**.** *Given f an entire function with genus h and order λ, we have the inequality*

$$h \leqslant \lambda \leqslant h + 1$$

From this theorem we get the useful corollary that if an entire function has finite order, it necessarily has finite genus.

Proof. Let f have finite genus h. Then f has a Weierstrauss factorization

$$f(z) = e^{g(z)} \prod_{n=1}^{\infty} (1 - \tfrac{z}{a_n}) exp\left(\sum_{j=1}^{h} \frac{z^j}{j a_n^j} \right)$$

From the definition, we know that $deg(g(z)) \leqslant h$ and that order of the product of the exponential factors cannot exceed the order of said factors since the contribution of the factor $\prod_{n=1}^{\infty} 1 - \frac{z}{a_n}$ to $|f(z)|$ becomes negligible relative to the exponential factors as $|z|$ becomes large.

It will now suffice to show that the order of the canonical product is $\leqslant h + 1$. The convergence of the canonical product implies that $\sum_n |a_n|^{-h-1} < \infty$.

If $\frac{z}{a_n} < 1$, we have

$$log\left| (1 - \tfrac{z}{a_n}) exp\left(\sum_{k=1}^{h} \frac{z^k}{k a_n^k} \right) \right| \leqslant \frac{|\frac{z}{a_n}|^{h+1}}{h+1} + \frac{|\frac{z}{a_n}|^{h+2}}{h+2} + \cdots \leqslant \frac{1}{h+1} \frac{|\frac{z}{a_n}|^{h+1}}{1 - |\frac{z}{a_n}|}$$

which means

$$(1 - |\tfrac{z}{a_n}|) log\left| (1 - \tfrac{z}{a_n}) exp\left(\sum_{k=1}^{h} \frac{z^k}{k a_n^k} \right) \right| \leqslant |\tfrac{z}{a_n}|^{h+1}$$

Given any $\frac{z}{a_n}$ and $h \geqslant 1$, we have

$$log\left| (1 - \tfrac{z}{a_n}) exp\left(\sum_{k=1}^{h} \frac{z^k}{k a_n^k} \right) \right| \leqslant log\left| (1 - \tfrac{z}{a_n}) exp\left(\sum_{k=1}^{h} \frac{z^k}{k a_n^k} \right) \right|$$

We now note that

$$log\left|(1-\tfrac{z}{a_n})exp\left(\sum_{k=1}^{h}\frac{z^k}{ka_n^k}\right)\right| \leqslant (2h+1)|\tfrac{z}{a_n}|^{h+1}$$

■

For brevity, we define

$$G(s) = \begin{cases} \left(\frac{|D|}{(2\pi)^2}\right)^{\frac{s}{2}}\Gamma(s) & \text{if } D<0 \\ \\ \left(\frac{|D|}{\pi^2}\right)^{\frac{s}{2}}\Gamma(\frac{s}{2})^2 & \text{if } D>0 \end{cases} \tag{27}$$

and note that as $|s| \to \infty$, we have

$$\xi_K(s) = \zeta_K(s)G(s)s(1-s) \ll e^{|s|^{1+\epsilon}}$$

which means that $\xi_K(s)$ has order one. By our earlier discussion, we know that there is a monomial $Ps+Q \in \mathbb{C}[s]$ satisfying

$$\xi_K(s) = e^{Ps+Q}\prod_{\rho}(1-\tfrac{s}{\rho})e^{\frac{s}{\rho}}$$

where the product is over all zeros ρ of $\xi_K(s)$ multiplied in pairs ρ, $\overline{\rho}$ and A and B are some unknown constants. The strict convergence requirements for the Weierstrauss factorization are such that

$$e^{Ps+Q}\prod_{\rho}(1-\tfrac{s}{\rho})e^{\frac{s}{\rho}} = e^{(P+\sum_{\rho}\frac{1}{\rho})s+Q}\prod_{\rho}(1-\tfrac{s}{\rho})e^{-\frac{s}{\rho}}$$

So setting $A = P + \sum_{\rho}\frac{1}{\rho}$ and $B = Q$, we have

$$\xi_K(s) = \zeta_K(s)G(s)s(1-s) = e^{As+B}\prod_{\rho}(1-\tfrac{s}{\rho})e^{-\frac{s}{\rho}} \tag{28}$$

Recall that the functional equation satisfies $\xi_K(s) = \xi_K(1-s)$, meaning we have

$$\xi_K(s) = \xi_K(1-s) \Rightarrow log(\xi_K(s)) = log(\xi_K(1-s))$$

Given $Re(s) > 1$, we differentiate each side of the above to get

$$\frac{\xi_K'(s)}{\xi_K(s)} = -\frac{\xi_K'(1-s)}{\xi_K(1-s)} \tag{29}$$

Differentiating the logarithm of the right side of Equation 28 gives

$$\frac{\xi_K'(s)}{\xi_K(s)} = \frac{\zeta_K'(s)}{\zeta_K(s)} + \frac{G'(s)}{G(s)} + \frac{1}{s} + \frac{1}{s-1} = A + \sum_{\rho}\frac{-\frac{1}{\rho}}{1-\frac{s}{\rho}} - \sum_{\rho}\frac{1}{\rho}$$

$$= A + \sum_{\rho} \frac{1}{s-\rho} - \sum_{\rho} \frac{1}{\rho} \tag{30}$$

but by Equation 29, this also equals

$$-\frac{\xi_K'(1-s)}{\xi_K(1-s)} = -A - \sum_{\rho} \frac{1}{1-s-\rho} + \sum_{\rho} \frac{1}{\rho}$$

Since every ρ has a corresponding $1-\rho$ (by the symmetry of ξ_K), the sum $\sum_\rho \frac{1}{1-s-\rho}$ is a rearrangement of $\sum\limits_{\rho} \frac{1}{\rho-s}$. This gives us

$$A + \sum_{\rho} \frac{1}{s-\rho} - \sum \frac{1}{\rho} = -A - \sum_{\rho} \frac{1}{\rho-s} + \sum \frac{1}{\rho} = -A + \sum_{\rho} \frac{1}{s-\rho} + \sum \frac{1}{\rho} \tag{31}$$

It follows that $A = \sum \frac{1}{\rho}$. Substituting this into Equation 30 yields

$$\frac{\zeta_K'(s)}{\zeta_K(s)} + \frac{G'(s)}{G(s)} + \frac{1}{s} + \frac{1}{s-1} = \sum \frac{1}{s-\rho}$$

Letting $s \in \mathbb{R}$ (and by our earlier requirement on the real part of s, $s > 1$), we have $\frac{\zeta_K'(s)}{\zeta_K(s)} < 0$ so

$$\frac{G'(s)}{G(s)} + \frac{1}{s} + \frac{1}{s-1} > \sum \frac{1}{s-\rho}$$

Writing $\rho = \beta + i\gamma$. If $\rho \in \mathbb{R}$ then $\frac{1}{s-\rho} = \frac{1}{s-\beta}$ with no conjugate to add. If $\rho \notin \mathbb{R}$ and $s \in \mathbb{R}$, adding conjugate pairs yields

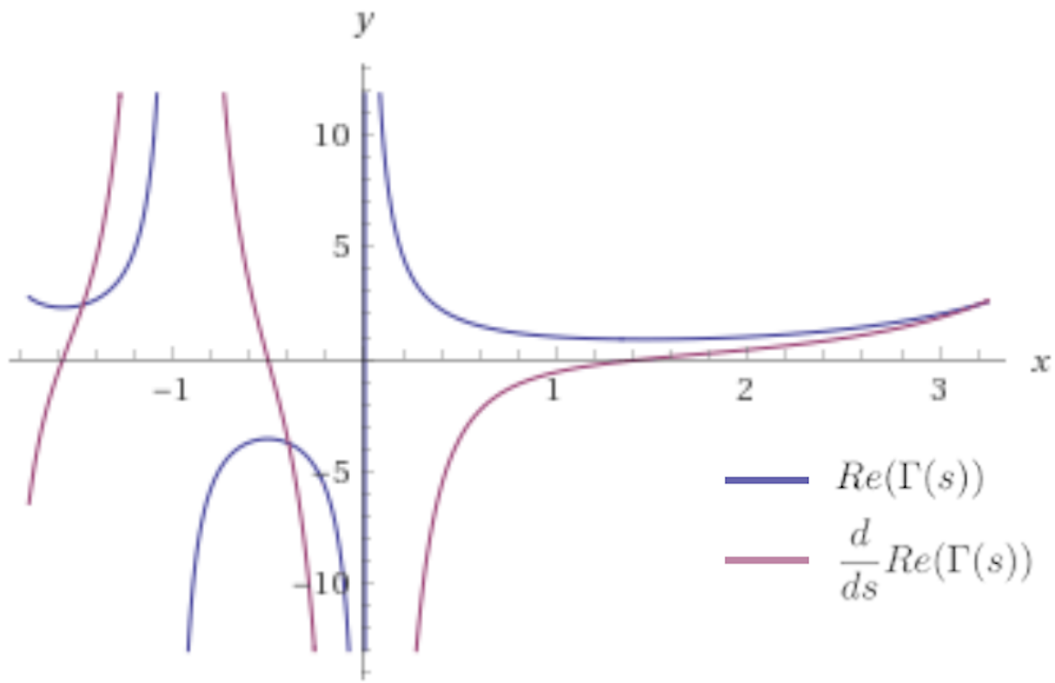
$$\frac{1}{s-\beta-i\gamma} + \frac{1}{s-\beta+i\gamma} = \frac{2(s-\beta)}{(s-\beta)^2+\gamma^2} > 0$$

so $\sum \frac{1}{s-\rho} > \sum\limits_{\rho\in\mathbb{R}} \frac{1}{s-\beta}$. From this we have

$$\frac{G'(s)}{G(s)} + \frac{1}{s} + \frac{1}{s+1} > \sum_{\beta\in\mathbb{R}} \frac{1}{s-\beta} \tag{32}$$

Examining the real plot of $\Gamma(s)$ and $\Gamma'(s)$, we observe that $\frac{\Gamma'(s)}{\Gamma(s)}$ is negative up until some point approximately half-way between 1 and 2. A numerical estimate for the positive real zero of $\Gamma'(s)$ seen in the plot is 1.46163, so by setting $1 < s < 1.4$, we certainly have $\frac{\Gamma'(s)}{\Gamma(s)} < 0$, and $\frac{\Gamma'(\frac{s}{2})}{2\Gamma(\frac{s}{2})} < 0$. Furthermore

$$\frac{G'(s)}{G(s)} = \begin{cases} \frac{1}{2}log|D| + \frac{\Gamma'(s)}{\Gamma(s)} - \frac{1}{2}log(4\pi^2) & \text{if } D < 0 \\ \\ \frac{1}{2}log|D| + \frac{\Gamma'(\frac{s}{2})}{\Gamma(\frac{s}{2})} - \frac{1}{2}log(\pi^2) & \text{if } D > 0 \end{cases}$$



so we know from *Equation* 32 that

$$\frac{1}{2}log|D| + \frac{1}{s} + \frac{1}{s-1} > \left[\frac{1}{2}log|D| + \frac{\Gamma'(s)}{\Gamma(s)}\right] + \frac{1}{s} + \frac{1}{s-1} > \sum_{\beta\in\mathbb{R}} \frac{1}{s-\beta} \tag{33}$$

Suppose we have a β such that $\frac{1}{logD} < \beta < 1$ for some constant c and satisfying $\zeta_K(\beta) = 0$. Setting $s = 1 + \frac{1}{4log|D|}$, we first observe that

$$\frac{1}{1-\beta} > \frac{1}{1 + \frac{1}{4log|D|} - \beta} = \frac{1}{s-\beta}$$

$$\frac{9}{2}log|D| + 1 > \frac{1}{2}log|D| + \frac{4log|D|}{1-4log|D|} + 4log|D| > \frac{1}{1-\beta} > 4log|D|.$$

Now suppose there is another zero β' such that $1 - \beta' < \frac{1}{clog|D|}$ for some fixed $c > 0$. Then we have

$$\frac{9}{2}log|D| + 1 > \frac{1}{1-\beta} + \frac{1}{1-\beta'} > (4+c)log|D|$$

which results in a contradiction when $c \geqslant \frac{1}{log|D|} + \frac{1}{2}$.

To be added: Twin Prime Conjecture from Infinite sequence of Siegel Zeros

An article of Heath-Brown discusses this.